

Problem Set 22

Problem 18.2.2: To first order, the amplitude $d_{2\ell m}(\infty)$ for the atom to be in the $|n = 2, \ell, m\rangle$ state is

$$d_{2\ell m}(\infty) = -\frac{i}{\hbar} \int_{-\infty}^{\infty} \langle 2\ell m | (e\mathcal{E}Z) | 100 \rangle e^{-t^2/\tau^2} e^{i\omega_{21}t} dt$$

since the potential $H^1(t) = -\vec{\mu} \cdot \vec{E} = eZ\mathcal{E}e^{-t^2/\tau^2}$. Here $\omega_{21} = (E_2 - E_1)/\hbar$, which from now on we'll just call ω . So we need to evaluate $\langle 2\ell m | Z | 100 \rangle$. Z is a component of a vector irreducible tensor operator, $Z = T_1^0$, so by angular momentum selection rules, only the $\langle 210 | Z | 100 \rangle$ matrix element is non-vanishing. This matrix element is easily evaluated using the wave functions for the $|210\rangle$ and $|100\rangle$ states, and that $Z = r \cos \theta$:

$$\begin{aligned} \langle 210 | Z | 100 \rangle &= \int_0^\infty r^2 dr \int d\Omega \left(\frac{1}{2^{5/2}\pi a_0^3} \right)^{1/2} \left(\frac{r}{a_0} e^{-r/2a_0} \cos \theta \right) (r \cos \theta) \left(\frac{1}{\pi a_0^3} \right)^{1/2} e^{-r/a_0} \\ &= 2\pi \left(\int_{-1}^1 d(\cos \theta) \cos^2 \theta \right) \cdot \left(\int_0^\infty dr r^4 e^{-3r/2a_0} \right) \cdot \frac{1}{2^{5/2}\pi a_0^4} \\ &= 2\pi \cdot \frac{2}{3} \cdot \left(\frac{2a_0}{3} \right)^5 \left(\int_0^\infty d\rho \rho^4 e^{-\rho} \right) \cdot \frac{1}{2^{5/2}\pi a_0^4} = 2^{15/2} \cdot 3^{-5} \cdot a_0. \end{aligned}$$

Therefore

$$\begin{aligned} d_{2\ell m}(\infty) &= \delta_{\ell 1} \delta_{m 0} \left(\frac{-i}{\hbar} \right) e\mathcal{E} \cdot 2^{15/2} \cdot 3^{-5} \cdot a_0 \int_{-\infty}^{\infty} e^{-t^2/\tau^2} e^{i\omega t} dt \\ &= \delta_{\ell 1} \delta_{m 0} \left(\frac{-ie\mathcal{E}}{\hbar} \right) 2^{15/2} \cdot 3^{-5} \cdot a_0 \sqrt{\pi\tau^2} e^{-\omega^2\tau^2/2}, \end{aligned}$$

and so the probability for the transition is

$$P(n=2) = \sum_{\ell, m} |d_{2\ell m}|^2 = \left(\frac{e\mathcal{E}}{\hbar} \right)^2 \left(\frac{2^{15} a_0^2}{3^{10}} \right) \pi \tau^2 e^{-\omega^2\tau^2/2}.$$

This answer does not depend on the electron spin: since S_z is conserved by H^0 and H^1 , there is still only a single final state it can go into.

Problem 18.2.4: The kinetic energy of the emitted electron is $16 \text{ keV} = \frac{1}{2}mv_e^2 = \frac{1}{2}mc^2(v_e/c)^2 = \frac{1}{2}(511\text{keV})(v_e/c)^2$, which implies that $v_e/c \sim 1/4$. Therefore the time, τ , for emission is $\tau \sim a_0/v_e = (a_0/c)/(v_e/c) \sim 4a_0/c$, since the typical size of the 1s electron orbit is $r \sim a_0$, the Bohr radius. In comparison, the characteristic time scale of the 1s electron, T , is $T \sim a_0/v_s = (a_0/c)/(v_s/c) = (a_0/c)/\alpha \sim 140a_0/c$, since

the typical velocity of an electron bound in the hydrogen atom is $v_s/c = \alpha$, the fine structure constant. Therefore, $T \gg \tau$, and the sudden approximation is appropriate.

In the sudden approximation, right after emission the 1s electron will be in the same state, the $|100\rangle_{(Z=1)}$ state of hydrogen. Therefore, the amplitude for the electron to be in the $|100\rangle_{(Z=2)}$ state of $(\text{He}^3)^+$ is given by the overlap

$$\begin{aligned} {}_{(Z=2)}\langle 100|100\rangle_{(Z=1)} &= \int_0^\infty r^2 dr \int d\Omega \left(\frac{Z^3}{\pi a_0^3} \right)^{1/2} e^{-rZ/a_0} \cdot \left(\frac{1}{\pi a_0^3} \right)^{1/2} e^{-r/a_0} \\ &= 4\pi \cdot \frac{2^{3/2}}{\pi a_0^3} \int_0^\infty r^2 dr e^{-3r/a_0} = \frac{2^{7/2}}{a_0^3} \left(\frac{a_0}{3} \right)^3 \int_0^\infty \rho^2 d\rho e^{-\rho} = 2^{9/2} \cdot 3^{-3}, \end{aligned}$$

where in the first line $Z = 2$ and I used the fact that under changing the hydrogen nucleus charge from 1 to Z , all that changes is $a_0 \rightarrow a_0/Z$.

Finally, ${}_{(Z=2)}\langle 16, 3, 0|100\rangle_{(Z=1)} = 0$ since $\ell = 3$ states are orthogonal to $\ell = 0$ ones (and the radial part, and therefore Z , does not affect this).